

Name: _____
Class: 12MT2__ or 12MTX__
Teacher: _____

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2012 AP4

YEAR 12 TRIAL HSC EXAMINATION

MATHEMATICS

Time allowed - 3 HOURS
(Plus 5 minutes reading time)

Directions to candidates

- Attempt all questions
- Approved calculators may be used.
- Standard Integral Tables are provided at the back of this paper.
- Write your name and class in the space provided at the top of this question paper

Section I - TOTAL MARKS 10

- To be answered on the removable answer grid at the back of the exam paper
- Allow about 15 minutes for this section

Section II - TOTAL MARKS 90

- All answers to be completed on the writing paper provided. Each question is to be commenced on a new page clearly marked Question 11, Question 12, etc on the top of the page. Write your name and class at the top of each page.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Allow about 2 hours and 45 minutes for this section

YOUR ANSWERS WILL BE COLLECTED IN ONE BUNDLE. THE MULTIPLE CHOICE SECTION I ON TOP AND THEN WRITTEN ANSWERS TO SECTION II AND THEN THE QUESTION PAPER.

SECTION I 10 MARKS

INSTRUCTIONS

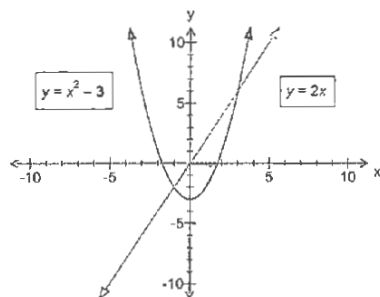
- Attempt all questions
- Allow about 15 minutes for this section
- Section I answers are to be completed on the multiple-choice answer sheet attached to the back of this question paper.
- Select the alternative A, B, C or D that best answers the question

1. Evaluate $\sqrt{6} + 7$ correct to 3 significant figures.
(A) 3.60
(B) 3.61
(C) 9.44
(D) 9.45
2. If the roots of the quadratic equation, $3x^2 - 9x + 5 = 0$, are α and β , then
(A) $\alpha\beta = -\frac{5}{3}$ and $\alpha + \beta = -3$
(B) $\alpha\beta = \frac{5}{3}$ and $\alpha + \beta = -3$
(C) $\alpha\beta = \frac{5}{3}$ and $\alpha + \beta = 3$
(D) $\alpha\beta = 3$ and $\alpha + \beta = \frac{5}{3}$
3. A parabola has as its focus (4,8) and directrix $y = -2$. Find the coordinates of the vertex.
(A) (4, 3)
(B) (8, 1)
(C) (2, 8)
(D) (1, 8)
4. Solve the pair of simultaneous equations.

$$\begin{array}{ll} 3x + y = 3 & (1) \\ 5x + 2y = 4 & (2) \end{array}$$

(A) $x = 2$ and $y = -3$
(B) $x = 2$ and $y = 3$
(C) $x = 4$ and $y = -3$
(D) $x = 4$ and $y = 3$

5. The shaded region shown in the diagram below is bounded by the functions, $y = x^2 - 3$ and $y = 2x$.



Which of the following is used to calculate the area of the shaded region

- (A) $\int_{-1}^3 (x^2 - 3 - 2x) dx$
 (B) $\int_{-1}^3 (x^2 - 3 + 2x) dx$
 (C) $\int_{-1}^3 (2x - x^2 - 3) dx$
 (D) $\int_{-1}^3 (2x - x^2 + 3) dx$

6. Evaluate

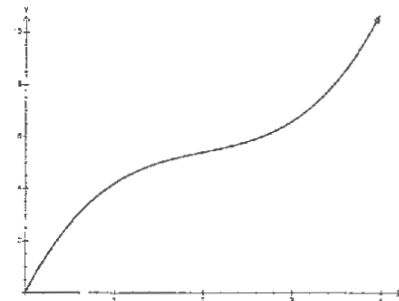
$$\sum_{r=3}^5 (2r + 4)$$

- (A) 24
 (B) 36
 (C) 50
 (D) 72
7. Solve $|2a + 7| > 9$
- (A) $a > -8, a < 1$
 (B) $a < -8, a > 1$
 (C) $a > -8$ and $a < 1$
 (D) $a < -8$ and $a > 1$

8. If $\ln g = 5$ and $\ln h = 9$ find the value of $\ln(gh^2)$

- (A) 6.003887067
 (B) 23
 (C) 86
 (D) 90

9. For the part of the function which is shown, which of the following properties are true?



- (A) $f'(x) > 0$ and $f''(x) > 0$
 (B) $f(x) \geq 0$ and $f'(x) > 0$
 (C) $f'(x) < 0$ and $f''(x) \neq 0$
 (D) $f(x) \geq 0$ and $f''(x) > 0$
10. The equation of any line that passes through the point of intersection of the lines, $3x + 2y + 1 = 0$ and $5x + y + 3 = 0$, can be written in the form

$$3x + 2y + 1 + k(5x + y + 3) = 0$$

Using this equation, an expression for the gradient of such a line is

- (A) $\frac{8k}{3}$
 (B) $\frac{-8k}{3}$
 (C) $\frac{3+5k}{2+k}$
 (D) $-\frac{3+5k}{2+k}$

SECTION II 90 MARKS

INSTRUCTIONS

- Answer all questions on the writing paper provided
- Allow about 2 hours and 45 minutes for this section
- Begin each question on a new page.
- Show all necessary working.

Question 11 (15 marks) BEGIN A NEW PAGE

- | | Marks |
|--|-------|
| (a) Simplify $\frac{x^2+10x+21}{3x^2-27}$ | 2 |
| (b) (i) Show that the sequence below is arithmetic
4, 9, 14, 19, ... | 1 |
| (ii) Hence find the sum of the first 100 terms of the series | 1 |
| (c) If $\frac{4}{\sqrt{2}+1} = a\sqrt{2} - b$, find the values of a and b . | 2 |
| (d) Find the equation of the tangent to the curve $y = x^2 - 4$ at the point where $x = 2$. Express your answer in gradient intercept form. | 3 |
| (e) The points A(2,0), B(6,2) and C(7,0) lie on a number plane. | |
| (i) Draw a neat labelled diagram that shows this information. | 1 |
| (ii) Find the gradient of side AB and the gradient of side BC. | 2 |
| (iii) Explain why $\triangle ABC$ is a right angled triangle. | 1 |
| (iv) Hence or otherwise, find the area of $\triangle ABC$. | 2 |

Question 12 (15 marks) BEGIN A NEW PAGE

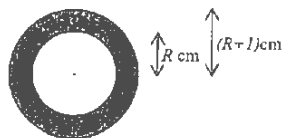
Marks

- | | | |
|-----|--|---|
| (a) | Evaluate $\int_1^3 (2x^3 - 9)dx$ | 2 |
| (b) | What is the exact value of $\cos \frac{\pi}{4} + \sin \frac{\pi}{3}$?
Answer as a single fraction. | 2 |
| (c) | Solve $5\cos\alpha = -4$ where $0 \leq \alpha \leq 2\pi$. Answer correct to 2 decimal places. | 2 |
| (d) | In the diagram below, the circle $x^2 + y^2 = 4$ and the line $y = \sqrt{3}x$ are shown. The point B(2,0) is one of the x-intercepts of the circle and the point A is the point of intersection of the line and circle, as shown on the diagram. | |
-
- | | | |
|-------|--|---|
| (i) | Show that $\angle AOB = \frac{\pi}{3}$. | 1 |
| (ii) | Hence find the area of sector AOB. | 1 |
| (iii) | Find the perimeter of sector AOB. | 1 |
| (e) | Differentiate | |
| i) | $\cos^4 x$ | 1 |
| ii) | $x^2 \tan x$ | 1 |
| (f) | Find the area bounded by the x-axis and one period of the curve $y = \sin x$. | 2 |
| (g) | Sketch the region $y > x^2$ and $y < 4$. | 2 |

Question 13 (15 marks) BEGIN A NEW PAGE

Marks

- (a) An annulus is a 2 dimensional shape formed when a small circle is removed from the centre of a larger circle. In the diagram the radius of the inner circle is $R\text{ cm}$ and the radius of the outer circle is 1 cm larger, that is $(R + 1)\text{ cm}$.



- (i) Show that the area, A , of the annulus can be given by $A = 2\pi R + \pi$. 2
- (ii) Find $\frac{dA}{dR}$ 1
- (iii) The radius of the inner circle is increasing at a rate of 2 cm/s . Find the rate of increase of the area. 1
- (b) During construction of a building the height of the building, is given by the equation $H = t - (t - 10)^2 + 100$, where H is measured in metres and t is the number of months since construction began. 2
- Determine the rate that the height is increasing after 3 months.
- (c) A particle is subjected to external forces, such that its displacement, x , is given by the equation $x = 4t^3 - 3t$, where displacement is given in metres and time in seconds.
- (i) Find expressions for the particle's velocity, $\dot{x}$, and acceleration, $\ddot{x}$. 2
- (ii) Find the acceleration of the object at $t = 5$ seconds. 1
- (iii) Find the time when the object is stationary and its position at this time. 2
- (iv) When is the particle accelerating in the positive direction? 1

Question 13 continues next page

Question 13 continued

- (d) The velocity, $\dot{x}\text{ m/s}$, of an object is given by the equation, $\dot{x} = 5t + 11$.
- (i) Find an equation for the displacement of the object if it has an initial displacement of -8 m . 2
- (ii) Hence find its displacement after 10 seconds. 1

Question 14 (15 marks) BEGIN A NEW PAGE

Marks

- (a) Differentiate with respect to x .
- (i) $y = e^{2x}$ 1
- (ii) $y = (e^{x^2} - 7)^4$ 2
- (iii) $y = \ln \sqrt{3x-9}$ 2
- (iv) $y = \log_4 5x$ 2
- (b) At the start of January 2008, Jin placed a number of fish in the dam on her farm. Over time the number of fish increases such that the rate of change of the population, $\frac{dP}{dt}$, is proportional to the population, P . That is $\frac{dP}{dt} = kP$, where k is a positive constant.
- (i) Show that an expression for the population can be written in the form $P = Ae^{kt}$. 1
- (ii) If Jin initially placed 10 fish in the dam, find A . 1
- (iii) At the end of 2009, there were 300 fish. Find the value of k , correct to 3 significant figures. 2
- (iv) Hence, estimate to the nearest 1000, the number of fish that will be in the dam at the end of 2012. 1
- (c) Evaluate $\int_1^2 \frac{10}{5x+8} dx$. Leave your answer as an exact value. 3

Question 15 (15 marks) BEGIN A NEW PAGE

Marks

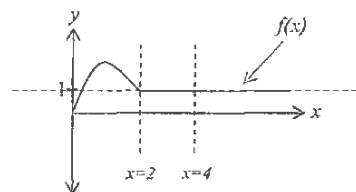
- (a) Trevor took out a loan of \$15000 to buy a car. Each month he is charged interest at 1.5% per month and makes a payment of \$M at the end of the month. A_n is the amount that Trevor still owes at the end of the n^{th} month.
- (i) By first finding expressions for A_1 and A_2 show that $A_3 = 15000 \times 1.015^3 - M(1.015^2 + 1.015 + 1)$. 2
- (ii) Hence write an expression for A_n . 1
- (iii) How much does Trevor need to pay each month to repay the loan in 36 months? 2
- (b) (i) Differentiate $\frac{14}{7x^2+2}$ 1
- (ii) Hence find $\int \frac{14x}{(7x^2+2)^2} dx$ 1
- (c) For the function $y = 2x^3 - 3x^2 - 12x + 18$, find
- (i) The first and second derivatives. 1
- (ii) Find any stationary points and determine their nature. 2
- (iii) Find the point of inflexion. 1
- (iv) Find the intercepts. 2
- (v) Hence sketch the function showing these features. 2

Question 16 (15 marks) BEGIN A NEW PAGE

Marks

- (a) Evaluate $\int_4^8 \frac{420}{x} dx$ by using Simpson's rule with 5 function values. 2

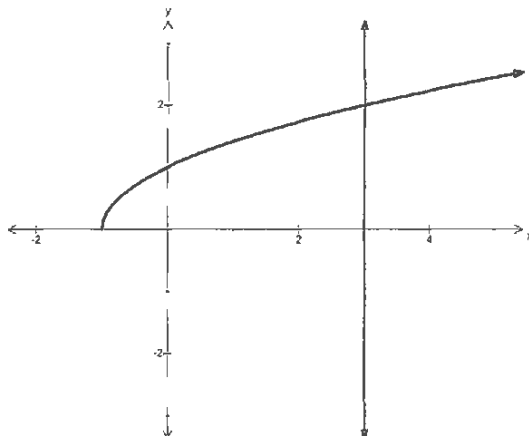
- (b) The diagram below shows the right hand side of an odd function, $f(x)$.



- (i) Copy or trace the diagram onto your writing paper and complete the left hand side of the graph. 1

- (ii) Hence evaluate $\int_{-2}^4 f(x) dx$. 1

- (c) The area enclosed by the curve $y = \sqrt{x+1}$, the x-axis and the line $x = 3$, is rotated around the x-axis. 2

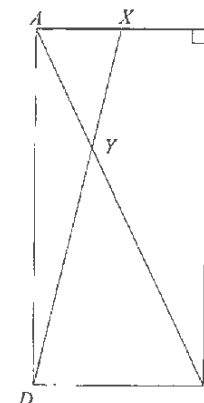


Find the exact volume of the solid formed.

Question 16 continues next page

Question 16 continued

- (d) In the diagram below, $ABCD$ is a rectangle. The point X is the midpoint of AB . The line AC meets XD at Y .



Not to scale

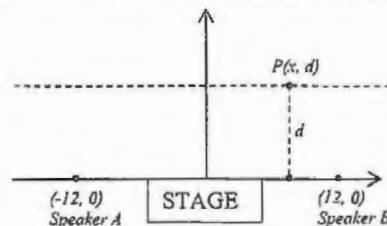
- (i) Prove that $\triangle AXY$ and $\triangle CDY$ are similar. 2

- (ii) Show that $CY = 2AY$. 1

Question 16 continues next page

Question 16 continued

- (e) At a music festival, the speakers at the front are placed on either side of the main stage and are 24 metres apart. The sound mixers are to be a distance of d metres from the stage (in a particular row).



It is known that the total sound, S , from these speakers at point $P(x, d)$ is:

$$S = \frac{100}{d^2 + (x + 12)^2} + \frac{100}{d^2 + (x - 12)^2}$$

- (i) Show that $\frac{dS}{dx} = -200 \frac{M}{Q}$, where 2

$$M = (x + 12)(d^2 + (x - 12)^2)^2 + (x - 12)(d^2 + (x + 12)^2)^2 \text{ and}$$

$$Q = (d^2 + (x + 12)^2)^2 (d^2 + (x - 12)^2)^2.$$

- (ii) By noting that: 2

$$M = 2x(x^2 + 144 + d^2 + 24\sqrt{144 + d^2})(x^2 + 144 + d^2 - 24\sqrt{144 + d^2}),$$

use $\frac{dS}{dx}$ to show that Mario the sound mixer, who moves along a row 20 metres from the stage, measures the sound to be at a maximum when in line with the centre of the stage.

- (iii) Another sound mixer (Aaron) decides it may be better to be closer to the stage. Aaron moves along a row 5 metres from the stage. 2

Describe how the sound level changes for Aaron as he moves along the row. Give clear reasons for your answer.

END OF PAPER

2012 Mathematics
2 Unit AP4

MC

1	D	6	B
2	C	7	B
3	A	8	B
4	A	9	B
5	D	10	D

$$\textcircled{11} \textcircled{a} \frac{x^2 + 10x + 21}{3x^2 - 27} = \frac{(x+7)(x+3)}{3(x+3)(x-3)} \quad \checkmark$$

$$= \frac{x+7}{3(x-3)} \quad \text{or} \quad \frac{x+7}{3x-9} \quad \checkmark$$

$$\textcircled{b} \textcircled{1} \frac{19-4}{3} = 5 \quad \checkmark \quad \text{or other working}$$

$$\textcircled{11} \textcircled{1} a=4, d=5, n=100$$

$$S_{100} = \frac{100}{2} (2 \times 4 + (100-1)5)$$

$$= 25150 \quad \checkmark$$

$$\textcircled{7} \textcircled{a} \frac{4 \times (\sqrt{2}-1)}{\sqrt{2}+1 \times (\sqrt{2}-1)} = \frac{4\sqrt{2}-4}{2-1} \quad \checkmark$$

$$= 4\sqrt{2}-4$$

$$a=4, b=4 \quad \checkmark$$

$$\textcircled{d} y = x^2 - 4$$

$$\frac{dy}{dx} = 2x \quad \checkmark$$

$$\text{at } x=2$$

$$m = 2 \times 2$$

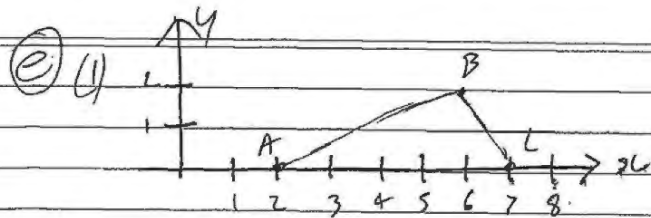
$$= 4$$

$$y = 2^2 - 4$$

$$= 0 \quad \checkmark$$

$$y - 0 = 4(x - 2)$$

$$y = 4x - 8 \quad \checkmark$$



point only required.

(ii) $m_{AB} = \frac{2-0}{6-2} = \frac{2}{4} = \frac{1}{2}$

$m_{BC} = \frac{0-2}{7-6} = \frac{-2}{1} = -2$

(iii) $m_{AB} \times m_{BC} = \frac{1}{2} \times -2$

$= -1$

$\therefore AB \perp BC$ so $\triangle ABC$ is right angled.

(iv) $AB = \sqrt{(2-0)^2 + (6-2)^2}$
 $= 2\sqrt{5}$

$BC = \sqrt{1^2 + 2^2}$
 $= \sqrt{5}$

Area $= \frac{1}{2} \times 2\sqrt{5} \times \sqrt{5}$
 $= 5u^2$

OR

Altitude through B = 2 units

$AC = 7-2 = 5$

Area $= \frac{1}{2} \times 2 \times 5$
 $= 5u^2$

② a $\int_1^3 (2x^3 - 9) dx$

$= \left[\frac{2x^4}{4} - 9x \right]_1^3$

$= \left[\frac{x^4}{2} - 9x \right]_1^3$

$= \frac{81}{2} - 27 - (1 - 9)$

$= 22$

(b) $\cos \frac{\pi}{4} + \sin \frac{\pi}{3}$

$= \frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2}$

$= \frac{\sqrt{2} + \sqrt{3}}{2}$

(c) $5 \cos \alpha = -4$

$\cos \alpha = -\frac{4}{5}$

$\begin{pmatrix} \sqrt{5} \\ \sqrt{1} \end{pmatrix} \begin{matrix} A \\ C \end{matrix}$

reference angle $= 0.6435 \dots$

$\alpha = \pi \pm 0.6435 \dots$

$= 2.4980915; 3.7850937 \dots$

$= 2.50, 3.79$

$$\begin{aligned} \textcircled{a} \text{ (i)} \quad \tan \theta &= m \\ \tan \theta &= \sqrt{3} \\ \theta &= \tan^{-1} \sqrt{3} \\ &= \frac{\pi}{3} \\ \angle AOB &= \frac{\pi}{3} \end{aligned}$$

✓ this line required or similar.

Note $\frac{\pi}{3}$ alone not accepted - Given in question.

$$\begin{aligned} \text{(ii)} \quad A &= \frac{r^2 \theta}{2} \\ &= \frac{2^2 \frac{\pi}{3}}{2} \\ &= \frac{2\pi}{3} \text{ units}^2 \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad 2 + 2 + 2 \times \frac{\pi}{3} \\ = 4 + \frac{2\pi}{3} \text{ units} \end{aligned}$$

ISE

$$\begin{aligned} \textcircled{a} \text{ (i)} \quad \frac{d}{dx} (\cos x)^4 \\ = 4(\cos^3 x) \times (-\sin x) \end{aligned}$$

✓

terms written in any order

$$\begin{aligned} \text{(ii)} \quad \frac{d}{dx} x^2 \tan x \\ = x^2 \sec^2 x + 2x \tan x \end{aligned}$$

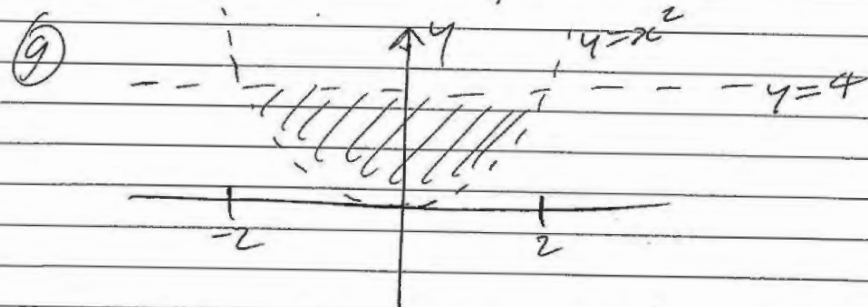
ISE

$$\begin{aligned} \textcircled{+} \quad A &= 2 \int_0^{\pi} \sin x \, dx \\ &= 2 [-\cos x]_0^{\pi} \\ &= 2 (1 - (-1)) \\ &= 4 \text{ units}^2 \end{aligned}$$

✓

Note. It is also correct to start the period at a different place

$$\text{eg } 2 \left| \int_{-\pi}^0 \sin x \, dx \right|$$



1 mark region shaded is enclosed by line & parabola.

2nd mark dotted boundary and at least one of the x coord of point of intersection.

Marking guidelines.

Question 12.

b) $\frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2}$ ✓

$\frac{2+\sqrt{6}}{2\sqrt{2}}$ or $\frac{\sqrt{2}+\sqrt{3}}{2}$ ✓

c) $\alpha = 2.50$ or 3.79 only ✓

Must show

2.50 , $2\pi - 2.50$ to get the 2nd mark.

g) 1 mark for the region

2nd mark for dotted boundary
and x intercept (5).

(13) a) (i) $A = \pi(R+1)^2 - \pi R^2$ ✓
 $= \pi(R^2 + 2R + 1) - \pi R^2$
 $= \pi R^2 + 2\pi R + \pi - \pi R^2$ ✓
 $= 2\pi R + \pi$

(ii) $\frac{dA}{dR} = 2\pi$ ✓

(iii) $\frac{dR}{dt} = 2$

$\frac{dA}{dt} = \frac{dR}{dt} \times \frac{dA}{dR}$

$= 2 \times 2\pi$

$= 4\pi \text{ cm}^2/\text{s}$ ✓

$$(b) H = t - (t-10)^2 + 100$$

$$\frac{dH}{dt} = 1 - 2(t-10) \quad \checkmark$$

$$\left[\begin{array}{l} \text{OR} \\ H = t - (t^2 - 20t + 100) + 100 \\ = -t^2 + 19t \\ \frac{dH}{dt} = -2t + 19 \end{array} \right]$$

$$\text{at } t = 3$$

$$\frac{dH}{dt} = 1 - 2(3-10)$$

$$= 1 - 2(-7) \\ = 15 \text{ m/month} \quad \checkmark$$

$$(c) x = 4t^3 - 3t$$

$$(i) \dot{x} = 12t^2 - 3$$

$$\ddot{x} = 24t$$

$$(ii) \text{ At } t = 5$$

$$\ddot{x} = 24 \times 5 \\ = 120 \text{ m/s}^2$$

$$(iii) 12t^2 - 3 = 0$$

$$4t^2 - 1 = 0$$

$$4t^2 = 1$$

$$t^2 = \frac{1}{4}$$

$$t = \pm \frac{1}{2} \text{ s} \quad \text{reject } -\frac{1}{2}$$

stationary at $t = \frac{1}{2} \text{ s}$ at -1 m

$$(d) (i) \dot{x} = 5t + 11$$

$$x = \int (5t + 11) dt$$

$$x = \frac{5t^2}{2} + 11t + C \quad \checkmark$$

$$\text{at } t = 0 \quad x = -8$$

$$-8 = 0 + 0 + C$$

$$C = -8$$

$$x = \frac{5t^2}{2} + 11t - 8 \quad \checkmark$$

$$(ii) \text{ at } t = 10$$

$$x = 352 \text{ m} \quad \checkmark$$

(14) (a) (i) $y = e^{2x}$

$y' = 2e^{2x}$

(ii) $y = (e^{x^2} - 7)^4$

$y' = 4(e^{x^2} - 7)^3 \cdot 2xe^x$
 $= 8xe^x(e^{x^2} - 7)^3$

(iii) $y = \ln(3x-9)^{1/2}$

$y' = \frac{1/2 \cdot 3(3x-9)^{-1/2}}{\sqrt{3x-9}}$
 $= \frac{3}{2(3x-9)}$

(iv) $y = \log_4 5x = \frac{\ln 5x}{\ln 4}$

$y' = \frac{1}{\ln 4} \cdot \frac{5}{5x}$
 $= \frac{1}{5x \ln 4}$ ISE

(b) (i) $P = Ae^{kt}$
 $\frac{dP}{dt} = kAe^{kt}$
 $= kP$

(ii) at $t=0$ $P=10$

$10 = Ae^0$

$A=10$

(iii) $P = 10e^{kt}$

at $t=2$ $P=300$

$300 = 10e^{2k}$

$30 = e^{2k}$

$2k = \ln 30$

$k = \frac{\ln 30}{2}$

$= 1.700598691$

$= 1.70$

(iv)

$t=5$ OR $P = 10e^{1.70 \times 5}$

$= 49147.6884$

$= 49000$

Using rounded
accept either.

$P = 10 \times e^{\frac{\ln 30}{2} \times 5}$

$= 49295$

$= 49000$

using exact

(c) $\int_1^2 \frac{10}{5x+8} dx = 2 \int_1^2 \frac{5}{5x+8} dx$

$= 2 \left[\ln(5x+8) \right]_1^2$

$= 2 (\ln 18 - \ln 13)$

$= 2 \ln \left(\frac{18}{13} \right)$

215

$$\begin{aligned} \text{a) (i)} \quad A_1 &= 15000(1.015) - M \\ A_2 &= (15000(1.015) - M)1.015 - M \\ &= 15000(1.015)^2 - M(1.015) - M \\ &= 15000(1.015)^2 - M(1.015 + 1) \quad \textcircled{1} \end{aligned}$$

$$\begin{aligned} A_3 &= (15000(1.015)^2 - M(1.015 + 1))1.015 - M \quad \textcircled{1} \\ &= 15000(1.015)^3 - M(1.015^2 + 1.015 + 1) \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad A_n &= 15000(1.015)^n - M(1 + 1.015 + \dots + 1.015^{n-1}) \quad \textcircled{1} \\ \text{(accepted)} \quad A_n &= 15000(1.015)^n - M \frac{(1.015^n - 1)}{1.015 - 1} \end{aligned}$$

(iii) Age 0 loan will be repaid in 36 months

$$0 = 15000(1.015)^{36} - M(1 + 1.015 + \dots + 1.015^{35})$$

$$0 = 15000(1.015)^{36} - M \frac{(1.015^{36} - 1)}{1.015 - 1}$$

$$M = \frac{15000(1.015)^{36} \times 0.015}{1.015^3 - 1}$$

$$M = 542.285933$$

$$M = \$542.29$$

AW1 for CFM →
incorrect formula with full
working shown.

Common mistake
 $0 = 15000 \times 1.015^{36} - M(1.015^3 - 1)$
 $M = \$562.31$ AW1 only.

$$\begin{aligned} \text{b) i) } \frac{d}{dx} (14(7x^2 + 2)^{-1}) &= -1 \times 14(7x^2 + 2)^{-2} \times 14x \quad \text{(marks not awarded for this line)} \\ &= -14 \times 14x \\ &= \frac{-196x}{(7x^2 + 2)^2} \quad \textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \int \frac{14x}{(7x^2 + 2)^2} dx &= \frac{-1}{14} \int \frac{-14 \times 14x}{(7x^2 + 2)^2} dx \\ &= \frac{-1}{14} \times \frac{14}{7x^2 + 2} + C \quad \text{(ignore C)} \\ &= -\frac{1}{7x^2 + 2} + C \quad \textcircled{1} \leftarrow \text{must simplify to} \end{aligned}$$

$$\text{c) } y = 2x^3 - 3x^2 - 12x + 18$$

$$\begin{aligned} \text{(i)} \quad \frac{dy}{dx} &= 6x^2 - 6x - 12 \\ \frac{d^2y}{dx^2} &= 12x - 6 \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{both for } \textcircled{1}$$

$$\begin{aligned} \text{(ii)} \quad \frac{dy}{dx} = 0 \quad 6x^2 - 6x - 12 &= 0 \\ x^2 - x - 2 &= 0 \end{aligned}$$

$$(x - 2)(x + 1) = 0$$

$$\therefore x = 2, -1$$

* must have x values
y values
+ correct tests for
min/max to get 2 marks

$$\text{When } x = 2 \quad \frac{d^2y}{dx^2} = 24 - 6 = 18 > 0 \therefore \text{min. } y = -2$$

$$x = -1 \quad \frac{d^2y}{dx^2} = -12 - 6 = -18 < 0 \therefore \text{max } y = 25$$

$\therefore$ minimum stat. pt at $(2, -2)$ and maximum stat. pt at $(-1, 25)$

$$\begin{aligned} \text{iii) } \frac{d^2y}{dx^2} = 0 \quad 12x - 6 &= 0 \\ x &= \frac{1}{2} \end{aligned} \quad \text{When } x = \frac{1}{2} \quad y = 2\left(\frac{1}{2}\right)^3 - 3\left(\frac{1}{2}\right)^2 - 12\left(\frac{1}{2}\right) + 18$$

$$y = 10\frac{3}{4}$$

Test

x	0	$\frac{1}{2}$	1
$\frac{d^2y}{dx^2}$	-6	0	6

$\therefore$ change in concavity

$\therefore$ point of inflexion at $(\frac{1}{2}, 11\frac{1}{4})$
 $(\frac{1}{2}, 23)$

If test not
shown → AW1

$$\text{iv) } x \text{ intercepts } y = 0 \quad 0 = 2x^3 - 3x^2 - 12x + 18$$

$$0 = x^2(2x - 3) - 6(2x - 3)$$

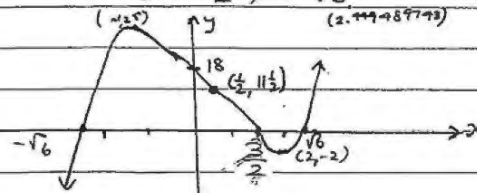
$$0 = (2x - 3)(x^2 - 6)$$

$$y \text{ intercept } x = 0$$

$$y = 18$$

$$\therefore x = \frac{3}{2}, \pm\sqrt{6} \quad \textcircled{1} \leftarrow \text{must have all 3}$$

v)



AS PER ORIGINAL
SOLUTIONS

* if pronounced horizontal point of
inflexion drawn → AW1 only

(16) @

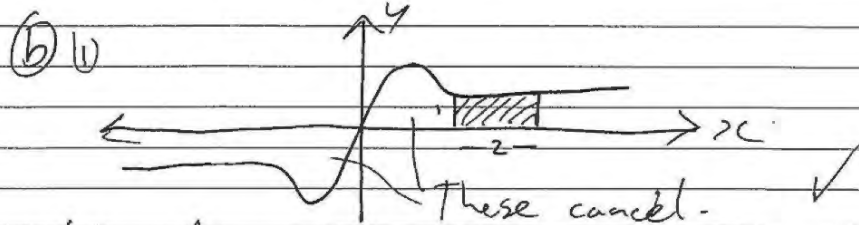
x	4	5	6	7	8
y	105	84	70	60	52.5

✓

$$A = \frac{1}{3} (105 + 52.5 + 4(84 + 60) + 2(70))$$

$$= \frac{1}{3} \times 873.5$$

$$\approx 291$$



(ii)

$$\int_{-2}^4 f(x) dx = 1 \times 2$$

$$= 2.$$

✓

(c)

$$V = \pi \int_{-1}^3 (x+1) dx$$

$$= \pi \left[\frac{x^2}{2} + x \right]_{-1}^3$$

$$= \pi \left(\frac{9}{2} + 3 - \left(\frac{1}{2} - 1 \right) \right)$$

$$= 8\pi u^3$$

ISE ✓

(d) (i) $\angle AXY = \angle CYD$ (vertically opposite $\angle$ s)
 $\angle XAY = \angle DCY$ (Alternate $\angle$ s $AB \parallel DC$)
 $\therefore \triangle AXY \sim \triangle CDY$ (equiangular)

(ii)

$$\frac{AX}{DC} = \frac{1}{2} = \frac{AY}{CY}$$

$$\therefore CY = 2AY$$

(iii)

$$S = \frac{100}{d^2 + (x+12)^2} + \frac{100}{d^2 + (x-12)^2}$$

$$S = 100[d^2 + (x+12)^2]^{-1} + 100[d^2 + (x-12)^2]^{-1}$$

$$\frac{dS}{dx} = -100[d^2 + (x+12)^2]^{-2} \times 2(x+12)$$

$$-100[d^2 + (x-12)^2]^{-2} \times 2(x-12)$$

$$= \frac{-200(x+12)}{[d^2 + (x+12)^2]^2} + \frac{-200(x-12)}{[d^2 + (x-12)^2]^2}$$

$$= \frac{-200[(x+12)[d^2 + (x-12)^2]^2 + (x-12)[d^2 + (x+12)^2]^2]}{[d^2 + (x+12)^2]^2 [d^2 + (x-12)^2]^2}$$

$$= -200 \frac{M}{Q}, \text{ as required}$$

(ii) The maximum sound that Mario experiences occurs when $\frac{dS}{dx} = 0$.

$$\therefore -200 \frac{M}{Q} = 0$$

i.e. when $M = 0$:

$$2x(x^2 + 144 + d^2 + 24\sqrt{144 + d^2})$$

$$\times (x^2 + 144 + d^2 - 24\sqrt{144 + d^2}) = 0$$

Putting $d = 20$, $\therefore d^2 = 400$

$$\text{i.e. } 2x(x^2 + 544 + 24\sqrt{544})(x^2 + 544 - 24\sqrt{544}) = 0$$

The only possibility is if $x = 0$ as both second and third expressions are positive.

When $x = 0$, the sound is minimum or maximum, however on checking: **MUST USE VALUES**

x	<	-1/2	0	1/2	>
S'	+	243772642	0	-243772642	-

$\therefore$ Sound level is maximum when $x = 0$.

(iii) For Aaron:

$$2x(x^2 + (144 + 25) + 24\sqrt{169})(x^2 + 169 - 24\sqrt{169}) = 0$$

$$x = 0, x^2 + 169 + 24\sqrt{169} = 0$$

$$\text{or } x^2 + 169 - 24\sqrt{169} = 0$$

$$\therefore x = 0 \text{ or } x^2 = 24 \times 13 - 169$$

$$x^2 = 143$$

$$x = \pm\sqrt{143}$$

Checking each of these:

x	<	$-\sqrt{143}$	>
S'	+	0	-

$\therefore$ maximum

$\therefore$ minimum

x	<	$+\sqrt{143}$	>
S'	+	0	-

$\therefore$ maximum

these are not always accepted! You should use values!

Hence, for Aaron the sound level will peak at two locations, when $x = \pm\sqrt{143} = \pm 11.96m$